Probe-Guided Steering of Worker-Bee Swarm Agents in a Simulated Hive

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Abstract

Alignment techniques for Multi-agent systems can be studied in controlled, low-stakes simulations before testing interventions in consequential systems. We use a synthetic animal society as such a testbed: 50 agents, each using the same Qwen3.5-0.8B model with a worker-bee LoRA adapter, operate a shared beehive through local observations, movement, feeding, cleaning, repair, and communication. A researcher-only vector records 14 normalized colony attributes after every action. Although the agents perform routine tasks, the aggregate hive health score falls from 0.820734 initially to roughly 0.795 after two simulated hours in unsteered runs. We train a linear, 14-output probe on model activations to predict the immediate change caused by an action, then derive and test two activation interventions: an equal combination of normalized attribute directions and an outcome-calibrated weighted combination. On held-out probe examples, the retrained probe reduces natural-action weighted scaled mean absolute error at layer 13 from 0.2230 to 0.1675 relative to an earlier probe. In paired two-hour simulations, learned steering exceeds the unsteered final score on five of six seeds, with mean gain 0.000719; equal steering succeeds on four of six, with mean gain 0.000319. The improvements are small and simulator-specific. The experiment suggests that a multi-output probe can support swarm interventions, however, its training-state coverage and action diversity are crucial to useful directions. The bounded setup can scale to larger synthetic swarms for further mitigation tests before large consequential multi-agent experiments.

1 Introduction

Autonomous agents can share resources and affect one another through local actions; collective outcomes may differ from the apparent intent of any single agent [2]. A small, inspectable society lets us measure these interactions and design mitigations without deploying agents in a real institution or ecosystem. We ask whether a representation-level intervention on one shared small model can improve the state of a colony controlled by many copies of that model.

Our hive contains independently acting adults with limited local information. They have no explicit colony score and cannot read or set its components. The experiment concerns *emergent performance of a bounded agent population*, not necessarily malicious bee goals. Its contributions are: (i) an action-logged hive with a hidden, calibrated state vector; (ii) a 14-output probe trained on actual per-action simulator changes in baseline and steered worlds across two simulated hours; and (iii) a paired comparison of equal and learned combinations of probe-column directions on six seeds. A seed is a number that initializes the simulation's randomness

2 Related Work

Apart's sprint identifies reproducible multi-agent testbeds and population-level oversight as open problems [1]. Multi-agent risk taxonomies include miscoordination and conflict [2]; GovSim studies LLM agents sharing a common resource [3]. Our study uses a task-rich physical colony rather than a strategic game: failure appears as drift in shared state despite plausible local behavior. Scalar-probe activation steering has been demonstrated on a small Qwen3-0.6B model for sycophancy [4]. Here the probe predicts a vector of action effects, and the columns of its weight matrix supply distinct candidate directions. Predictive fit alone does not establish that injecting probe weights is beneficial.

Four bee models informed selected abstractions, without their code or parameters being reproduced. TaskSelSim models decentralized task choice and nectar flow among foragers, storers, and nurses [5]. ApisRAM motivates

individual agents in spatial and environmental conditions with food transport and thermoregulation [6]. SIMBACA motivates inspectable self-organized task allocation and visualization [7]. Bee++ motivates worker-duty transitions, brood, stores, and finite forage resources [8]. Our 50-bee, two-hour graph simulation is an engineering testbed, not a validated biological synthesis of these projects.

3 Methods

3.1 Hive and agent interface

The Python world has 19 connected hive compartments and four landscape nodes. The initial population is one queen, 49 workers, and 16 non-acting brood. Workers have flexible roles; only the queen can lay eggs. Adults carry bounded honey, nectar, pollen, or water. Actions include movement, foraging, eating, transfer, brood feeding, cleaning, grooming, comb repair, heating, cooling, fanning, communicating, guarding, resting, role switching, and egg laying. Python enforces preconditions and resource and energy costs; a proposed action can be rejected. Time advances in 15-minute rounds, with one decision per living adult per round. For GPU throughput, up to 16 agents decide from a frozen world view; their actions then execute sequentially, so later actions can encounter depleted resources.

Each bee sees its own state, current and adjacent cells, nearby bees and brood, weather, private memory, local messages, and a feasible-action menu. It cannot see the global hive vector, score, or inventory. The observer computes $v_t \in [0,1]^{14}$ after each action. Its coordinates are adult survival, adult vitality, adult energy sufficiency, carbohydrate reserve, pollen reserve, water reserve, initial-brood survival, brood nutrition, thermal suitability, humidity suitability, hygiene, structural integrity, pathogen suppression, and queen vitality. Higher is better for each. The reported score is the *unweighted* mean $S_t = 14^{-1} \sum_{j=1}^{14} v_{tj}$. These are engineering indices, not empirical probabilities. Immediate action changes and elapsed-time background changes are logged separately (Appendix A).

3.2 Small-model policy and supervised data

We generated 11,900 synthetic local-observation/action examples: 700 for each of 17 actions, split 9,520/1,190/1,190 into training, validation, and test. A deterministic scenario teacher chose physically executable actions; these are plausible demonstrations, not expert labels or an optimal policy. A seeded, action-stratified half of the training split (4,760 rows) trained a rank-16 LoRA adapter on Qwen3.5-0.8B for two epochs using Unsloth and completion-only loss [9, 10]. One-epoch continuation on 1,485 failure-heavy and feasible-menu examples produced the adapter used here; 200 separate examples validated that stage. A constrained local menu and engine checks enforce feasible decisions. The 50 agents share model weights but have distinct observations, locations, memories, and action histories. Steering modifies inference activations, not LoRA weights.

3.3 Vector probe

For an action-conditioned example, a neutral forward pass yields mean response-token activation $h_\ell \in \mathbb{R}^d$ at each text decoder layer. The target is the observed *immediate* step change $\Delta v \in \mathbb{R}^{14}$, including zeros for unaffected attributes. A multi-output ridge probe fits $\widehat{\Delta v} = h_\ell W_\ell + b_\ell$ after standardizing features and scaling targets from training data only. Rows with rare changes receive capped weights. A separate change-mask classifier was fitted, but steering uses the dense ridge weights; sparse-mask F1 was not evaluated in this final retraining.

Training uses 1,600 natural actions from baseline and earlier steered worlds on seeds 90–91, covering both hours, plus 1,023 cloned-world targeted or controlled actions. Controlled cases separate pollen from brood nutrition and hygiene from pathogen suppression. Layer choice uses 563 examples from seed 110; an independent probe report uses 1,134 examples from seeds 111–112 (800 natural and 334 targeted). Layers 13 and 14 were selected. Error is weighted scaled mean absolute error (MAE), with nonzero target entries weighted four times unchanged entries; *lower is better.*

3.4 Steering and paired evaluation

Only eight probe columns had usable change signals. At each selected layer, column $W_{\ell j}$ becomes a unit vector $u_{\ell j} = W_{\ell j} / \|W_{\ell j}\|_2$. We screen each direction with both signs on matched frozen batches and choose the sign with

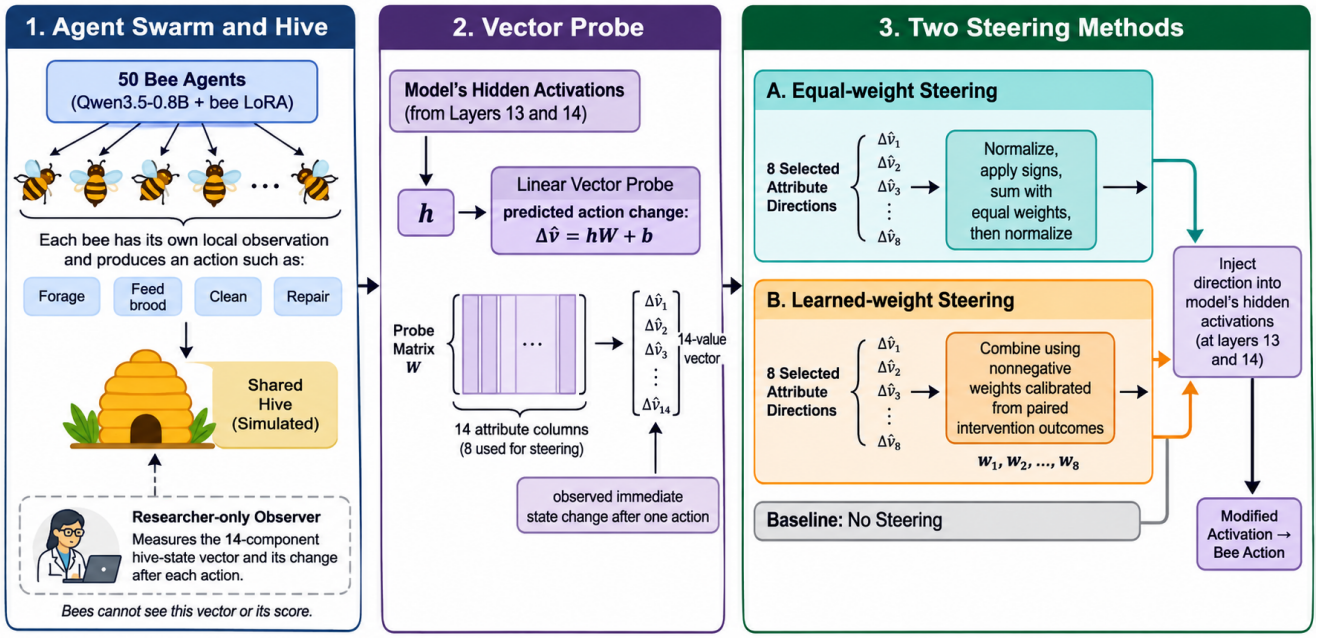


Figure 1: Schematic overview of the simulated bee swarm, vector probe, and steering methods.

Table 1: Probe prediction error on seeds 111–112. Weighted scaled MAE is lower when prediction is better; these are regression errors, not classification accuracies. Each hour/environment slice has equal weight in the reported mean.

Layer	Action set	Examples	Earlier probe	Retrained probe
13	Natural	800	0.2230	**0.1675**
13	Targeted	334	0.4880	**0.0726**
14	Natural	800	0.2050	**0.1659**
14	Targeted	334	0.4599	**0.0788**

better post-batch hive-score change. The equal arm normalizes the sum of signed unit columns; the learned arm normalizes their weighted sum. Nonnegative weights come from a softmax of measured paired batch effects, shrunk 25% toward uniform. They are *outcome-calibrated*, not trained by backpropagation through the simulator. Both arms inject at every token in layers 13 and 14, at strength 0.045 times a fixed training-activation norm per layer, divided by $\sqrt{2}$. Signs, weights, and strength were chosen on screen worlds before final hive seeds.

Final comparisons run baseline, equal steering, and learned steering for two simulated hours, 400 actions per arm, on seeds 114–115 and then 100–103. No probe refit or direction recalibration occurs on 100–103. Seeds 114–115 were fresh for probe fitting and calibration; seeds 100–103 had appeared in earlier experiments using *other* directions, so they are not wholly untouched study seeds. Each seed's three arms share initial conditions and simulator randomness. Figure 1 summarizes the agent, probe, and steering setup.

4 Results

The revised data improved prediction on both layers and action sets (Table 1). Layer-13 natural-action error fell about 25%. Hour-two error nevertheless remained above hour-one error (0.1995–0.2264 versus 0.1193–0.1248 at layer 13). The larger targeted-action gain reflects deliberate enrichment of rare effects and should not be extrapolated to natural action frequencies. Learned steering exceeds baseline on five of six seeds; equal steering does so on four. The learned paired mean gain is 0.000719 on a 0–1 score, with sample standard deviation 0.000615 across six paired effects. The largest learned gain is 0.001545 (seed 100); seed 101 worsens by 0.000082. Direction choice was exploratory, and six partly reused seeds do not justify a population-level significance claim. All arms start at 0.820734 and end lower. Steering reduces the observed decline relative to baseline. Background drift is substantial,

Table 2: Final hive score after two simulated hours; higher is better. Bold marks the highest score in each seed, including ties. Δ is the paired score minus baseline.

Seed	Baseline	Equal	Δ equal	Learned	Δ learned
100	0.795404	0.796777	+0.001372	**0.796949**	+0.001545
101	**0.796590**	0.796412	−0.000177	0.796507	−0.000082
102	0.798808	0.799194	+0.000386	**0.799339**	+0.000530
103	0.792145	0.791805	−0.000340	**0.793506**	+0.001362
114	0.794336	0.794497	+0.000161	**0.794782**	+0.000446
115	0.795238	**0.795749**	+0.000511	**0.795749**	+0.000511
Mean	0.795420	0.795739	+0.000319	**0.796139**	+0.000719

Table 3: Hive score at one and two simulated hours. Higher is better; bold marks the highest score within each seed and hour. Each hour contains 200 decisions.

	One hour			Two hours		
Seed	Baseline	Equal	Learned	Baseline	Equal	Learned
100	0.809546	0.810656	**0.810761**	0.795404	0.796777	**0.796949**
101	**0.810693**	0.810452	0.810547	**0.796590**	0.796412	0.796507
102	0.812795	**0.813320**	0.812727	0.798808	0.799194	**0.799339**
103	0.807470	0.806902	**0.808900**	0.792145	0.791805	**0.793506**
114	0.809206	0.809238	**0.809434**	0.794336	0.794497	**0.794782**
115	0.809234	**0.810316**	**0.810316**	0.795238	**0.795749**	**0.795749**
Mean	0.809824	0.810147	**0.810447**	0.795420	0.795739	**0.796139**

and no no-agent or optimal-policy counterfactual was run. The learned mean advantage over baseline is 0.000623 at one hour and 0.000719 at two hours (Table 3; Figure 2). On seeds 100–103, the learned arm chose 27 brood-feeding actions versus 16 in baseline. Mean brood nutrition was 0.019825 higher, while pollen reserve was 0.004406 lower; steering changes task allocation and resource tradeoffs, not every attribute in a favorable direction. The 100–103 transcript contains 4,800 valid action snapshots and reproduces earlier unsteered baseline vectors on those seeds. All final arms had zero schema-invalid actions.

5 Discussion and Limitations

The probe predicts immediate local effects, while the target outcome is a two-hour collective trajectory under feedback. Changing an action alters the observations later bees receive. Fitting only common, early, unsteered actions previously left important columns weak or entangled. Adding steered hour-two environments and targeted separations improved held-out prediction and produced useful directions across more seeds. Still, prediction quality does not prove a direction is beneficial; signs and combination weights required a separate intervention screen.

The simulator fixes its own physics and welfare definition. Its score equally weights all 14 attributes, including near-constant coordinates; biological parameters were not calibrated to field data. Fifty adults cannot represent a real colony's demography. Synthetic teacher labels and a feasible-action menu constrain behavior. Final differences are small, seed 101 illustrates nonuniform benefit, and seeds 100–103 are not entirely untouched. We did not test long horizons, other model families, adversarial agents, or transfer to human multi-agent systems. A no-action control, more untouched seeds, varied strength, and different teacher policies are natural next tests. The required ethics appendix gives further limits.

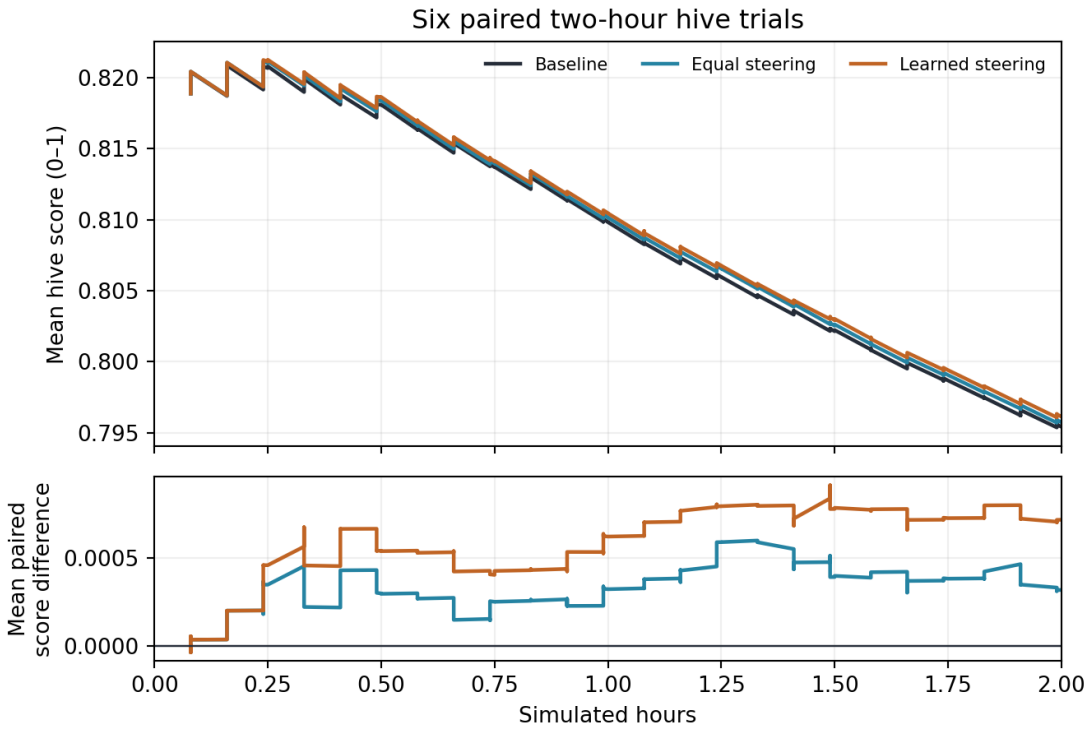


Figure 2: Mean hive score across six matched seeds over two simulated hours. Each line averages post-action snapshots at the same decision index; separation is small relative to common downward drift. Higher is better.

6 Conclusion

A bounded synthetic hive exposes population-level outcomes of many locally acting LLM agents while preserving an action-level audit trail. A 14-output probe trained on diverse state-action changes supported a small positive learned-steering effect on five of six paired two-hour runs. This is an exploratory simulator finding, not proof of robust swarm alignment. Successful learned steering depended strongly on the probe's training environment: adding both steered and unsteered hour-one and hour-two states, plus rare-effect examples, was necessary to obtain useful multi-seed directions in this project.

Code and Data. Repository: `https://github.com/Faithful777/apart_swarm_alignment`. The companion code directory contains the simulator, dataset generators, LoRA training notebooks, final probe notebooks, and result tables. Model and dataset provenance, including pre-sprint work, is documented in its README.

References

[1] Apart Research. 2026. *Swarm Alignment Hackathon: Guidelines and Resources.* `https://apartresearch.com/sprints/swarm-alignment-2026`.

[2] Hammond et al. 2025. *Multi-Agent Risks from Advanced AI.* arXiv:2502.14143. `https://arxiv.org/abs/2502.14143`.

[3] Piatti et al. 2024. *Cooperate or Collapse: Emergence of Sustainable Cooperation in a Society of LLM Agents.* arXiv:2404.16698. `https://arxiv.org/abs/2404.16698`.

[4] TensorSlay. 2025. *Activation Steering With Mean Response Probes: A Case Study in Suppressing Sycophancy in Language Models During TTC.* Hugging Face Community Article. `https://huggingface.co/blog/TensorSlay/activation-steering-with-mean-response-probes`.

[5] Schmickl, T., and K. Crailsheim. 2008. *TaskSelSim: A Model of the Self-Organization of the Division of Labour in Honeybees. Mathematical and Computer Modelling of Dynamical Systems.* `https://doi.org/10.1080/13873950701846662`.

[6] Duan, X., D. Wallis, F. Hatjina, N. Simon-Delso, A. Bruun Jensen, and C. J. Topping. 2022. *ApisRAM Formal Model Description*. EFSA Supporting Publication EN-7184. https://doi.org/10.2903/sp.efsa.2022.EN-7184.

[7] Alves, T. 2018–2021. *SIMBACA: Visualisation and Interactions with a Virtual Honey Bee Colony*. Lab-STICC doctoral project description. https://labsticc.fr/en/teams/inuit/past-phd-thesis/phd-thomas-alves.

[8] Betti, M., J. LeClair, L. M. Wahl, and M. Zamir. 2017. *Bee++: An Object-Oriented, Agent-Based Simulator for Honey Bee Colonies*. *Insects* 8(1):31. https://doi.org/10.3390/insects8010031.

[9] Qwen Team. 2026. *Qwen3.5-0.8B Model Card*. https://huggingface.co/Qwen/Qwen3.5-0.8B.

[10] Unsloth. 2026. *Qwen3.5 Fine-tuning Guide*. https://unsloth.ai/docs/models/qwen3.5/fine-tune.

A Simulator and measurement detail

The observer clips explicit engineering formulas to $[0, 1]$: adult and initial-brood survival fractions; living-adult health and energy fractions; food reserves relative to fixed targets; brood nutrition from reference-cohort hunger and health; closeness to 35 °C and 60% relative humidity; low debris and pathogen load; mean comb integrity; and queen health. Exact definitions and update equations are in `beehive_sim.py` and `MODEL_DESIGN.md`. Every action proposal records a post-action vector, simulated timestamp, immediate vector change, and any background change from elapsed time. The tabulated score is always the arithmetic mean of 14 components, never the engine's optional researcher-weighted score.

Source trajectories on seeds 90–91 were replayed with zero vector error, and local observations were reconstructed without new model decisions. Targeted actions were executed in cloned worlds and labeled with actual immediate vector changes. The later evaluation on seeds 100–103 reconstructed two missing activation-norm scalars from the original 2,623 probe-training examples, without fitting new weights. Both final transcripts were audited for bounded coordinates, sequential events, matching score arithmetic, and equality of immediate plus background change with total step change.

B Limitations and Dual-Use / Ethical Considerations

Hive physics, teacher labels, and 0–1 calibration are synthetic and unvalidated against field colonies. Improving this score may exploit simulator simplifications and does not imply real bee welfare. Agents share model weights and face neither strategic adversaries nor realistic communication channels. Steering settings were chosen after exploratory screens. The 5/6 successful runs count is descriptive, not a generalization guarantee. Multi-agent steering and monitoring could make coordinated agent populations more capable or harder to detect in other contexts. Here agents operate within a bounded simulator with no external tools, and action-level logs permit inspection. Use beyond such a sandbox requires separate safety analysis and independent validation.

C Prior Work and Reproducibility

The beehive simulator, synthetic SFT generator, and initial LoRA training were developed before the November 6–8 sprint. The menu policy and continuation training were prepared by November 5. The final retrained probe, direction screen, and six-seed evaluations were completed during the sprint period. Provenance is marked in the companion README. All displayed numbers come from saved Kaggle reports or action transcripts.

D LLM Usage Statement

We used an OpenAI coding assistant (Codex) to help develop code and deploy experiments, inspect saved results, and draft this report. The methods infographic in Figure 1 was generated with ChatGPT (OpenAI) from our study description. The numerical tables and results figure were checked against saved action transcripts and experiment reports. We edited and reviewed the report and infographic before submission.